

Final Testing Report — FR-20: Xem sản phẩm (Mobile)

Assignment: HW02 — AI-First Domain Testing & Boundary Value Analysis
Feature ID: FR-20
Feature: Xem sản phẩm (Mobile)
Date: 2026-07-07
AI Tool: Antigra²vity (Gemini 3.5 Flash)
SUT: EShop Mobile App on Android Emulator (emulator-5554) — <http://localhost:8081>

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1. Feature Overview

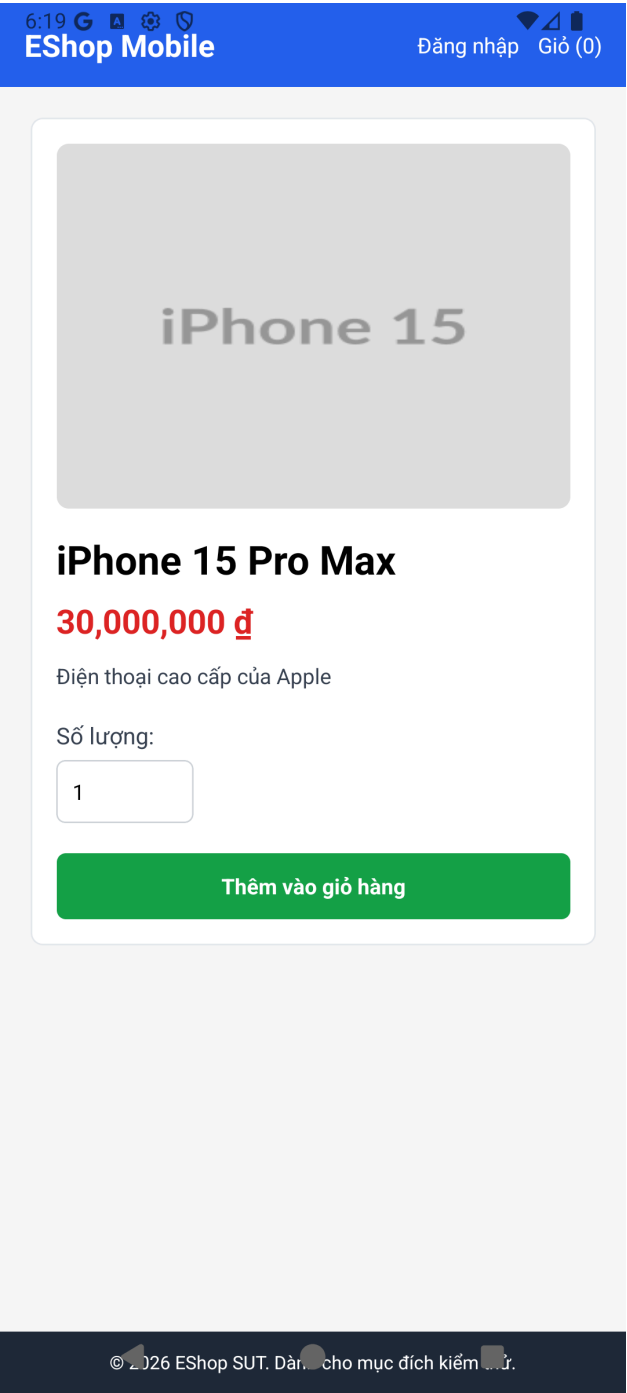
Item	Detail
Feature	Xem sản phẩm (Mobile)
Actor	Guest / Registered User
Entry Point	Home screen catalog list -> Tap "Xem chi tiết" on any product card
API Endpoints	GET /api/products (Retrieve product catalog) GET /api/products/:id (Retrieve single product details)
Front-end States	home (Product List), productDetail (Detail View), cart (Shopping Cart View)
Add to Cart logic	Handled purely in-memory on client React Native state. (No API call is triggered).

2. Testing Environment

Component	URL / Details	Status
Expo Metro Server	http://localhost:8081	 Running

Component	URL / Details	Status
Backend API Server	http://localhost:3000	✓ Running
Android Emulator	emulator-5554 (Android 16)	✓ Connected & Running
Automation Control	Android Debug Bridge (ADB) coordinates/inputs	✓ Operational

Evidence: [ENV-01-product-screen.png](#)



3. Domain Testing Summary

Methodology

Technique: **Equivalence Partitioning / Domain Testing**
Artifacts: DT-01 → DT-02 → DT-03 → DT-04 (each reviewed via REVIEW-01)

Test Cases

TC ID	Scenario	Partitions Covered	Expected Result	Actual Result (SUT)	Status
TC-DT-001	View Odd Product ID + valid quantity (Nominal)	PID-P1, QTY-P1	Detail view loads price: 30,000,000 đ . Success alert popped; cart count increments.	Rendered successfully. Success alert popped; unique product count added.	✓ PASS
TC-DT-002	View Even Product ID + default quantity	PID-P2, QTY-P1	Detail view loads price: 28,000,000 đ (string price parsed to numeric formatting).	Price parsed and rendered as 28,000,000 đ . Success alert popped.	✓ PASS
TC-DT-003	View non-existent product details	PID-P3	SUT returns empty object {} . Client displays blank state message.	Backend returns {} and SUT renders "Sản phẩm không tồn tại (Lỗi trắng trang do data rỗng)" .	✓ PASS
TC-DT-004	Add to cart with zero quantity	PID-P1, QTY-P2	Spec: Blocked with alert. SUT Actual: Silently coerces to 1 .	Silently normalizes to 1; success alert shown.	✗ FAIL (Spec) / ✓ PASS (Bug verified)
TC-DT-005	Add to cart with negative quantity	PID-P1, QTY-P3	Spec: Blocked with alert. SUT Actual: Silently coerces to 1 .	Silently normalizes to 1; success alert shown.	✗ FAIL (Spec) / ✓ PASS (Bug verified)

TC ID	Scenario	Partitions Covered	Expected Result	Actual Result (SUT)	Status
TC-DT-006	Add to cart with decimal quantity	PID-P1, QTY-P4	Spec: Blocked or rounded. SUT Actual: Truncated to int.	Decimal truncated (2.5 -> 2); success alert shown.	✗ FAIL (Spec) / ✓ PASS (Bug verified)
TC-DT-007	Add to cart with non-numeric quantity	PID-P1, QTY-P5	Spec: Blocked with validation warning.	Silently normalizes to 1; success alert shown.	✗ FAIL (Spec) / ✓ PASS (Bug verified)
TC-DT-008	Add to cart with exceptionally large quantity	PID-P1, QTY-P6	Spec: Blocked with upper limit warning.	Accepts input, total calculations render in scientific notation and wrap UI.	✗ FAIL (Spec) / ✓ PASS (Bug verified)

4. Boundary Value Analysis Summary

Methodology

Technique: **Boundary Value Analysis (BVA)**

The variable `quantity` has a clear expected lower boundary of `1` (positive integer). No upper bound is explicitly specified in requirements.

Test Cases

TC ID	Boundary Point	Test Value	Expected Result	Actual Result (SUT)	Status
TC-BVA-001	Minimum - 1 (min-1)	0	✗ Rejected — show error alert	✗ Silently normalizes to 1 and adds to cart	FAIL (Coercion Flaw)
TC-BVA-002	Minimum (min)	1	✓ Added to cart with quantity 1	✓ Added to cart with quantity 1	PASS
TC-BVA-003	Minimum + 1 (min+1)	2	✓ Added to cart with quantity 2	✓ Added to cart with quantity 2	PASS

TC ID	Boundary Point	Test Value	Expected Result	Actual Result (SUT)	Status
TC-BVA-004	Nominal	5	✅ Added to cart with quantity 5	✅ Added to cart with quantity 5	PASS

5. Execution Results

Summary

Technique	Total TCs	PASS (Meets Spec)	FAIL (Specification / Logic Gaps)
Domain Testing	8	3	5
BVA Testing	4	3	1
Total	12	6	6

6. Bug Reports

BUG-FR20-001 — Medium Severity (Validation Flaw / Input Coercion)

- **Title:** Mobile quantity input lacks user-facing validation feedback and silently coerces invalid inputs (zero, negative, alphanumeric, or empty) to 1 (or truncates float decimals to integer).
- **File:** bugs/FR20/BUG-001.md

BUG-FR20-002 — Low Severity (Visual Layout / Calculation Overflow)

- **Title:** Quantity field lacks an upper bound check, allowing exceptionally large numbers (e.g. 99999999999) to overflow total prices into scientific notation and break the cart/checkout UI layouts.
- **File:** bugs/FR20/BUG-002.md

7. AI Gap Analysis

Full analysis in tests/FR20/GAP-01-gap-analysis.md .

Gaps Identified

1. **Cart Header Counter (G-01):** The nav header count (e.g., `Giò (1)`) represents the number of unique product entities in the cart array (`cart.length`), not the total accumulated product quantities.
2. **Missing Input Checking (G-02, G-03):** Decimals are truncated using `parseInt(value, 10)` , and zeros/negatives/letters default silently to 1 inside `normalizeQuantity` instead of showing error feedback.
3. **No Max Constraint (G-04):** No quantity capping is implemented, risking total price visual overflows on mobile screens.

8. Conclusion

Feature Status: ⚠️ OPERATIONAL BUT FLACCID VALIDATION

The feature successfully implements basic viewing and catalog loading logic. However, the quantity input field lacks rigorous **input validation**. Instead of rejecting invalid quantities (zero, negative, text, or extremely large values) with proper validation messages, the client silently normalizes them. This causes layout issues during overflow and leads to poor UX where users are misled about what was actually added to their shopping cart.

Recommended Fixes

- 1. **Implement Quantity Validators:** Update the quantity text field handler or the "Thêm vào giỏ hàng" trigger in `App.js` to reject values ≤ 0 , non-numeric values, or floats with a visible warning alert, rather than relying on silent default coercions.
- 2. **Define an Upper Limit:** Cap the quantity selector at a maximum value (e.g. 99 items per product) to prevent visual styling breakage in calculation text fields.

Artifacts Index

Artifact	Path
ENV-01 Report	ENV-01-environment-report.md
DT-01 Feature Understanding	DT-01-feature-understanding.md
REVIEW-01 of DT-01	REVIEW-01-of-DT-01.md
DT-02 Domain Identification	DT-02-domain-identification.md
REVIEW-01 of DT-02	REVIEW-01-of-DT-02.md
DT-03 Domain Partitioning	DT-03-domain-partitioning.md
REVIEW-01 of DT-03	REVIEW-01-of-DT-03.md
DT-04 Test Cases	DT-04-test-cases.md
EXEC-01 (Domain Execution)	execution.md
BVA-01 Boundary Analysis	BVA-01-boundary-analysis.md
EXEC-01 (BVA Execution)	execution-bva.md
GAP-01 Gap Analysis	GAP-01-gap-analysis.md
BUG-001 (Missing Quantity Validation)	BUG-001.md
BUG-002 (Large Quantity Overflow)	BUG-002.md